CIGNA MEDICAL COVERAGE POLICIES - MUSCULOSKELETAL

CMM-311: Knee Replacement/ Arthroplasty

Effective Date: August 04, 2026

Instructions for use

The following coverage policy applies to health benefit plans administered by Cigna. Coverage policies are intended to provide guidance in interpreting certain standard Cigna benefit plans and are used by medical directors and other health care professionals in making medical necessity and other coverage determinations. Please note the terms of a customer's particular benefit plan document may differ significantly from the standard benefit plans upon which these coverage policies are based. For example, a customer's benefit plan document may contain a specific exclusion related to a topic addressed in a coverage policy.

In the event of a conflict, a customer's benefit plan document always supersedes the information in the coverage policy. In the absence of federal or state coverage mandates, benefits are ultimately determined by the terms of the applicable benefit plan document. Coverage determinations in each specific instance require consideration of:

1. The terms of the applicable benefit plan document in effect on the date of service
2. Any applicable laws and regulations
3. Any relevant collateral source materials including coverage policies
4. The specific facts of the particular situation

Coverage policies relate exclusively to the administration of health benefit plans. Coverage policies are not recommendations for treatment and should never be used as treatment guidelines.

This evidence-based medical coverage policy has been developed by EviCore, Inc. Some information in this coverage policy may not apply to all benefit plans administered by Cigna.

These guidelines include procedures EviCore does not review for Cigna. Please refer to the **Cigna CPT code list** for the current list of high-tech imaging procedures that EviCore reviews for Cigna.

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Definitions

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Definitions

Focal Articular Resurfacing	a surgical procedure that resurfaces a localized area of damaged knee cartilage and underlying subchondral bone using a metal or metal-polyethylene implant, typically in a focal region of the knee (e.g., femoral condyle).
Kellgren-Lawrence Grading System	a radiographic grading system describing osteoarthritic changes to the tibial-femoral joint of the knee. When used, the radiographic findings on plain x-rays are typically reported within one of the following categories: • Grade I: doubtful narrowing of joint space and possible osteophytic lipping • Grade II: definite osteophytes and possible narrowing of joint space • Grade III: moderate multiple osteophytes, definite narrowing of joint space, some sclerosis, and possible deformity of bone contour • Grade IV: large osteophytes, marked narrowing of joint space, severe sclerosis, and definite deformity of bone contour
Knee Arthroplasty	a surgical procedure aimed at restoring knee function due to damaged bone/cartilage by replacing, resurfacing, or realigning the articular surfaces of the knee joint (partial or total). Arthroplasty is a broad term that includes various techniques, such as partial resurfacing, compartmental replacement, or total joint replacement.
Knee Replacement	a specific type of knee arthroplasty involving prosthetic replacement of knee compartments. When the medial and lateral compartments are replaced (with or without patellar resurfacing), it is termed total knee replacement/total knee arthroplasty; when one compartment is replaced, it is termed partial or unicompartmental knee replacement.

Modified Outerbridge Classification	a system developed for judging articular cartilage injury to the knee. This system allows delineation of varying areas of chondral pathology based on the qualitative appearance of the cartilage surface as viewed on MRI, and can assist in identifying those injuries that are suitable for repair techniques. The characterization of cartilage in this system is as follows: • Grade I: softening with swelling • Grade II: fragmentation and fissuring <1cm^2 • Grade III: fragmentation and fissuring >1cm^2 • Grade IV: subchondral bone exposed
Non-Surgical Management (with regard to the treatment of lower extremity joint pain)	any provider-directed non-surgical treatment which has been demonstrated in the scientific literature as efficacious and/or is considered reasonable care in the treatment of lower extremity joint pain. The types of treatment involved can include, but are not limited to, the following: relative rest/activity modification; weight loss; supervised physiotherapy modalities and therapeutic exercises; prescription and non-prescription medications; assistive devices; and/or intra-articular injections.
Outerbridge Classification	a system that has been developed for judging articular cartilage injury to the knee. This system allows delineation of varying areas of chondral pathology based on the qualitative appearance of the cartilage surface as viewed by direct visualization intraoperatively, and can assist in identifying those injuries that are suitable for repair techniques. The characterization of cartilage in this system is as follows: • Grade I: softening with swelling • Grade II: fragmentation and fissuring <1cm^2 • Grade III: fragmentation and fissuring >1cm^2 • Grade IV: subchondral bone exposed

Partial Knee Replacement	a surgical procedure that replaces the damaged bone or cartilage in one compartment (unicompartmental) of the knee with prosthetic components. In the unicompartmental knee replacement (UKA), both articular surfaces (the femoral surface and the corresponding tibial surface) of the corresponding compartment (e.g., medial, lateral, or patellofemoral) are replaced while preserving the other compartments of the knee.
Prosthesis	an artificial device used to replace a structural element within a joint to improve and enhance function.
Revision of Knee Replacement (Partial or Total)	surgical reconstruction or replacement due to failure or complications of previous knee replacement.
Total Knee Replacement	a surgical procedure that replaces all compartments of the knee (medial, lateral, and patellofemoral) with prosthetic components.

Replacement/Arthroplasty,

General Guidelines

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General Guidelines

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Application of Guideline

- The determination of medical necessity for the performance of knee replacement (partial or total) is always made on a case-by-case basis.
- For indications and non-indications of trochleoplasty using CPT® 27442 for trochlear dysplasia in individuals with patellar instability, refer to **Procedures for Patellofemoral Conditions** in **CMM-312: Knee Surgery - Arthroscopic and Open Procedures**.

Health Equity Considerations

Health equity is the highest level of health for all individuals; health inequity is the avoidable difference in health status or distribution of health resources due to the social conditions in which individuals are born, grow, live, work, and age. Social determinants of health are the conditions in the environment that affect a wide range of health, functioning, and quality of life outcomes and risks. Examples include the following: safe housing, transportation, and neighborhoods; racism, discrimination, and violence; education, job opportunities, and income; access to nutritious foods and physical activity opportunities; access to clean air and water; and language and literacy skills.

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Partial Knee Replacement Indications

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Partial Knee Replacement - Unicompartmental (Medial, Lateral, or Patellofemoral)

Partial knee replacement (medial, lateral, or patellofemoral) is considered **medically necessary** when ALL of the following criteria have been met:

- Imaging or arthroscopic findings show EITHER of the following:
  - severe unicompartmental (medial, lateral, or patellofemoral) osteoarthritis as evidenced by ANY of the following:
    - Kellgren-Lawrence Grade IV radiographic findings
    - Outerbridge Classification Grade IV arthroscopic findings AND not a candidate for joint sparing procedure
    - Modified Outerbridge Classification Grade IV MRI findings AND not a candidate for joint sparing procedure
  - unicompartmental avascular necrosis (AVN) of the femoral condyles and/or proximal tibia
- Physical exam demonstrates BOTH of the following findings:
  - knee stability
  - knee arc of motion (full extension to full flexion) >90º
- Symptoms include BOTH of the following:
  - function-limiting pain at short distances (e.g., walking less than one-quarter mile, limiting activity to two [2] city blocks, the equivalent to walking the length of a shopping mall) for at least three (3) months duration
  - loss of knee function which interferes with the ability to carry out age-appropriate activities of daily living and/or demands of employment
- Failure of provider-directed non-surgical management for at least three (3) months' duration
  - **Criteria exception**: Provider-directed non-surgical management may be inappropriate. The medical record must clearly document why provider-directed non-surgical management is not appropriate.
  - **Note**: It is incumbent on the surgeon to preoperatively optimize reasonably modifiable medical and behavioral health comorbidities.

Patellofemoral Unicompartmental Replacement Following a Total Knee Replacement

Patellofemoral unicompartmental replacement is considered **medically necessary** when ALL of the following criteria have been met:

- Procedure is performed to manage protracted anterior knee pain and/or mechanical symptoms attributed to the patellofemoral joint following a total knee replacement, during which patellar replacement was NOT performed at the time of the index knee replacement
- Physical exam demonstrates BOTH of the following findings:
  - intact, stable ligaments
  - knee arc of motion (full extension to full flexion) >90º
- Symptoms include BOTH of the following:
  - function-limiting pain at short distances (e.g., walking less than one-quarter mile, limiting activity to two [2] city blocks, the equivalent to walking the length of a shopping mall)
  - loss of knee function which interferes with the ability to carry out age-appropriate activities of daily living and/or demands of employment
- Failure of provider-directed non-surgical management for at least three (3) months' duration
  - **Criteria exception**: Provider-directed non-surgical management may be inappropriate. The medical record must clearly document why provider-directed non-surgical management is not appropriate.
  - **Note**: It is incumbent on the surgeon to preoperatively optimize reasonably modifiable medical and behavioral health comorbidities.

Partial Knee Replacement Non-Indications

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Not Medically Necessary

Partial knee replacement (medial, lateral, or patellofemoral) is considered **not medically necessary** for ANY other indication, condition, or when ANY of the following are present:

- when unicompartmental replacement is to be performed of the medial or lateral compartment:
  - Grade IV patellofemoral joint osteoarthritis involving the lateral patella facet and/or lateral trochlea as evidenced by EITHER of the following:
    - Kellgren-Lawrence Grade IV radiographic findings
    - Outerbridge Classification Grade IV arthroscopic findings
- when unicompartmental replacement is to be performed of the patellofemoral compartment:
  - Grade III or IV medial or lateral compartment osteoarthritis as evidenced by EITHER of the following:
    - Kellgren-Lawrence Grade III or IV radiographic findings
    - Outerbridge Classification Grade III or IV arthroscopic findings
  - patellar malalignment syndrome as evidenced by EITHER of the following:
    - increased Q angle (15° in males or 20° in females)
    - TT-TG (tibial tuberosity-trochlear groove) distance of 20mm
- tibial or femoral shaft deformity
- radiographic evidence of medial or lateral subluxation
- flexion contracture >15°
- valgus deformity >20°
- varus deformity >15°
- varus deformity ≥10° with sagittal tibiofemoral subluxation ≥6mm (consistent with an anterior cruciate ligament (ACL) deficient osteoarthritic knee)
- Charcot joint
- inflammatory arthropathy
- active local or systemic infection
- vascular insufficiency (defined as ankle brachial index of <0.5), significant muscular

Experimental, Investigational, or Unproven (EIU)

Based on lack of scientific evidence of efficacy and safety, the following are considered **experimental, investigational, or unproven**:

- bicompartmental knee arthroplasty (modular or monolithic/nonmodular)
- bi-unicompartmental knee arthroplasty
- focal resurfacing of a single knee joint defect (e.g., Arthrosurface® femoral condyle implant)
- unicompartmental free-floating (un-fixed) interpositional device (e.g., UniSpacer®)

Total Knee Replacement

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Total Knee Replacement Indications

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Total knee replacement is considered **medically necessary** for ANY of the following conditions when ALL of the associated criteria have been met:

Fracture of Distal Femur

- Imaging shows a fracture of the distal femur (trochlea, condyles).
- Conservative management or surgical fixation is not considered a reasonable option.

Osteoarthritis (OA) or Avascular Necrosis (AVN)

- Imaging or arthroscopic findings show EITHER of the following:
  - severe unicompartmental (medial, lateral, or patellofemoral), bicompartmental, or tricompartmental osteoarthritis as evidenced by ANY of the following:
    - Kellgren-Lawrence Grade III or IV radiographic findings
    - Outerbridge Classification Grade IV arthroscopic findings AND not a candidate for joint sparing procedure
    - Modified Outerbridge Classification Grade IV MRI findings AND not a candidate for joint sparing procedure
  - avascular necrosis of the femoral condyles and/or proximal tibia
- Symptoms include BOTH of the following:
  - function-limiting pain at short distances (e.g., walking less than one-quarter mile, limiting activity to two [2] city blocks, the equivalent to walking the length of a shopping mall) for at least three (3) months duration
  - loss of knee function which interferes with the ability to carry out age-appropriate activities of daily living and/or demands of employment
- Failure of provider-directed non-surgical management for at least three (3) months' duration
  - **Criteria exception**: Provider-directed non-surgical management may be inappropriate. The medical record must clearly document why provider-directed non-surgical management is not appropriate.
  - **Note**: It is incumbent on the surgeon to preoperatively optimize reasonably modifiable medical and behavioral health comorbidities.

Total Knee Replacement Non-Indications

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Not Medically Necessary

Total knee replacement is considered **not medically necessary** for ANY other indication, condition, or when ANY of the following are present:

- joint instability (due to a lack of collateral ligament integrity) that is not amendable to surgical correction (e.g., specialized implant, constrained implant, or a hinge implant)
- >30° of fixed varus deformity that is not amendable to surgical correction
- >30° of fixed valgus deformity that is not amendable to surgical correction
- individuals undergoing dialysis
- active local or systemic infection
- vascular insufficiency (defined as ankle brachial index of <0.5), significant muscular atrophy of the leg, or neuromuscular disease that is severe enough to compromise implant stability or post-operative recovery

Experimental, Investigational, or Unproven

Based on lack of scientific evidence of efficacy and safety, the following are considered **experimental, investigational, or unproven**:

- bicompartmental knee arthroplasty (modular or monolithic/nonmodular)
- bi-unicompartmental knee arthroplasty

Revision of Knee Replacement

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Revision of Knee Replacement (Partial or Total) Indications

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Revision of knee replacement includes any of the following: revision of a total knee replacement; revision of a unicompartmental (medial, lateral, or patellofemoral) replacement to another unicompartmental (medial, lateral, or patellofemoral) replacement; or revision of a unicompartmental (medial, lateral, or patellofemoral) replacement to a total knee replacement.

- Revision of knee replacement is considered **medically necessary** for an individual who has previously undergone a partial or total knee replacement when ANY of the following post-operative conditions are present:
  - patellar fracture or dislocation
  - aseptic loosening
  - periprosthetic infection
  - periprosthetic fracture
  - implant fracture or component failure
  - post-operative stiffness for more than 12 weeks when BOTH of the following criteria have been met:
    - manipulation is deemed unsafe by provider
    - components are well-positioned, well-fixed, and appropriately-sized
  - post-operative stiffness due to component sizing or positioning
  - instability of the knee
  - clinically significant, symptomatic limb malalignment due to existing component position
  - function-limiting pain at short distances (e.g., walking less than one-quarter mile, limiting activity to two [2] city blocks, the equivalent to walking the length of a shopping mall), that is unexplained, greater than six (6) months' duration, and is unresponsive to provider-directed non-surgical management
  - If **revising from a partial** (unicompartmental) knee replacement **to a total** joint replacement: Kellgren-Lawrence Grade IV radiographic findings in the non-replaced compartments (medial, lateral, or patellofemoral)

Revision of Knee Replacement (Partial or Total) Non-Indications

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Not Medically Necessary

Revision of knee replacement is considered **not medically necessary** for ANY other indication or condition.

Isolated Polyethylene Liner Exchange (IPE) Indications

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Isolated polyethylene liner exchange (IPE) is considered **medically necessary** for ANY of the following post-operative conditions when ALL of the associated criteria have been met:

- eccentric polyethylene wear and osteolysis with imaging studies confirming BOTH of the following findings:
  - progressive osteolysis
  - well-fixed implants in acceptable position
- catastrophic polyethylene failure, (includes post fracture, locking mechanism failure, or severe polyethylene wear) without component loosening or malalignment
- acute post-operative joint infection (periprosthetic or hematogenous) with well-fixed implants
- stiffness following total knee replacement (flexion contracture >15° with flexion limited to <90°) with BOTH of the following:
  - individual presents later than three (3) months from the index replacement procedure
  - persistent restricted range of motion despite BOTH of the following treatments:
    - physical therapy
    - manipulation under anesthesia
- instability without component malrotation or malalignment

Isolated Polyethylene Liner Exchange (IPE) Non-Indications

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Isolated polyethylene liner exchange (IPE) is considered **not medically necessary** for ANY other indication or condition.

Arthroscopic or Open Abrasion Arthroplasty Procedures of the Patella (without prosthesis), Femoral Condyles, or Tibial Plateau

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Arthroscopic or Open Abrasion Arthroplasty Procedures of the Patella (without prosthesis), Femoral Condyles, or Tibial Plateau Non-Indications

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Not Medically Necessary

Arthroscopic or open abrasion arthroplasty procedures of the knee tibial plateau(s), patella, and/or femoral condyle(s) (with or without debridement and partial synovectomy) are considered **not medically necessary** for the treatment of symptomatic knee osteoarthritis.

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Codes (CMM-311)

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The inclusion of any code in this table does not imply that the code is under management or requires prior authorization. Refer to the applicable health plan for management details. Prior authorization of a code listed in this table is not a guarantee of payment. The Certificate of Coverage or Evidence of Coverage policy outlines the terms and conditions of the member's health insurance policy.

Code	Code Description/Definition
27438	Arthroplasty, patella; with prosthesis
27440	Arthroplasty, knee, tibial plateau
27441	Arthroplasty, knee, tibial plateau; with debridement and partial synovectomy
27443	Arthroplasty, femoral condyles or tibial plateau(s), knee; with debridement and partial synovectomy
27445	Arthroplasty, knee, hinge prosthesis (e.g., Walldius type)
27446	Arthroplasty, knee, condyle and plateau; medial OR lateral compartment
27447	Arthroplasty, knee, condyle and plateau; medical AND lateral compartments with or without patella resurfacing (total knee Arthroplasty)
27486	Revision of total knee Arthroplasty, with or without allograft; 1 component
27487	Revision of total knee Arthroplasty, with or without allograft; femoral and entire tibial component

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